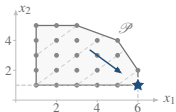
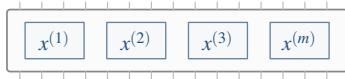


Mixed-integer programs



(a) GPU-native primal heuristics

Parallel candidates $\mathcal{C}$



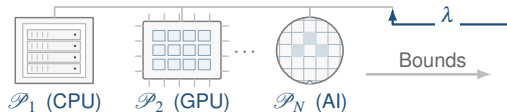
$$x^+ \in \arg \min_{x \in \mathcal{C} \cap \mathcal{X}} f(x)$$

Select a feasible candidate

National freight networks



(b) Distributed decomposition

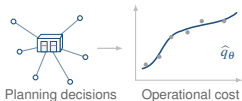


Dual coordination

$$L(\lambda) \leq f^* \leq f(\bar{x})$$

$\bar{x} \in \mathcal{X}$

Neural embedded models



(c) Learning and optimization on device

GPU / AI accelerator

Train
 $\theta \leftarrow \text{learn}(\mathcal{D})$

$\hat{q}_\theta$

Optimize

$$x \in \mathcal{X}, \quad y = \hat{q}_\theta(x)$$